VISAYAS STATE UNIVERSITY
Faculty of Computing
Department of Computer Science and Technology
Visca, Baybay City, Leyte 6521

Technical Evaluation Instrument

Product Quality of the Baybayani System, based on ISO/IEC 25010:2023
For developers, testers and system architects

Research project	Farm to Table: A Management Information System for Farmers and Food Business Establishments in Baybay City, Leyte (Project Code DCST.0824-0825)
System evaluated	Baybayani (web application and Android app)
Project Leader	Mr. Nelson H. Tejara
Study/Component Leaders	Mr. Jomari Joseph A. Barrera, Dr. Jonah Flor O. Maaghop, Mr. Rodney M. Maniego Jr., Dr. Jude B. Rola, and Dr. Magdalene C. Unajan

Part I. Informed Consent and Data Privacy Notice

Faculty researchers of the Department of Computer Science and Technology of Visayas State University are evaluating the product quality of Baybayani, the farm-to-table ordering system of the research project named above. We are asking you, as a technical stakeholder, to rate the system against the nine product quality characteristics of ISO/IEC 25010:2023, based on your own review, testing or analysis.

- **Voluntary.** Taking part is voluntary. You may skip any item or stop at any time.
- **Time.** Rating takes about 30 to 45 minutes after you have reviewed or tested the system.
- **Confidentiality.** Your name appears only on this page, which is filed separately from your ratings. Ratings are identified only by a respondent number and are reported as group summaries.
- **Data privacy.** Your information is collected, stored and used only for this evaluation, in line with the Data Privacy Act of 2012 (Republic Act No. 10173). Only the researchers can access it, and it is disposed of securely once the project is completed.
- **Confidential material.** Access to the source code, builds and test accounts is given only for this evaluation. Please do not copy, share or keep them after the evaluation.

For questions about this evaluation, approach any of the researchers at the Department of Computer Science and Technology, Visayas State University.

☐ I have read and understood this notice, and I agree to take part.

_____ _____

Signature over printed name Date

Part II. Evaluator Profile

Tick (✓) the box that applies.

1. Primary role	☐ Software developer ☐ Software tester / QA ☐ System architect ☐ Other: _____
2. Affiliation	☐ Academe (faculty) ☐ Industry ☐ Government ☐ Other: _____
3. Years of professional experience in the role	☐ Less than 1 ☐ 1–3 ☐ 4–6 ☐ 7–10 ☐ More than 10
4. Involvement in developing Baybayani	☐ None ☐ Consulted or reviewed during development ☐ Member of the development team
5. Familiarity with the technologies used (N = none, B = basic, P = proficient)	React and TypeScript ☐ N ☐ B ☐ P Supabase / PostgreSQL (RLS, Realtime) ☐ N ☐ B ☐ P Capacitor / Android ☐ N ☐ B ☐ P Web security and testing practice ☐ N ☐ B ☐ P
6. How you evaluated the system (tick all that apply)	☐ Hands-on use of the web app ☐ Hands-on use of the Android app ☐ Source code review ☐ Database schema and access-policy review ☐ Functional or exploratory testing ☐ Performance or load testing ☐ Security testing ☐ Architecture or documentation review

Evaluation environment (filled in by the evaluator or the researchers)

Version evaluated (Git commit or release)	
Web URL tested	
Android device and version tested	
Browsers and devices tested	
Test accounts used	☐ Customer ☐ Administrator
Evaluation date(s)	

Part III. Product Quality Evaluation

Directions. Rate each statement according to the evidence you gathered from your review or testing. Shade or tick (✓) one circle per item. Where a rating is based on a specific observation (a test case, a file, a screen, a log), note it in the observation box below the characteristic, especially for ratings of 2 or lower.

Rating	Code	Description	Mean range
5	SA	Strongly Agree	4.21–5.00
4	A	Agree	3.41–4.20
3	N	Neutral	2.61–3.40
2	D	Disagree	1.81–2.60
1	SD	Strongly Disagree	1.00–1.80

The mean range is used by the researchers to interpret the results.

1. Functional Suitability

The degree to which the system provides functions that meet the stated and implied needs of its users.

No.	Statement	5 SA	4 A	3 N	2 D	1 SD
1.1 Functional completeness						
1	The customer functions cover the intended tasks: sign-up with valid-ID upload, browsing and search, cart, checkout, order tracking and cancellation, chat, announcements and profile settings.	◯	◯	◯	◯	◯
2	The administrator functions cover the intended tasks: products and variants, stock movements, order processing (including walk-in orders), user approval, announcements, store hours, dashboard and report export.	◯	◯	◯	◯	◯
1.2 Functional correctness						
3	Computed values are correct: cart and order totals, retail and wholesale pricing, and dashboard revenue and order figures.	◯	◯	◯	◯	◯
4	Stock levels stay consistent with recorded movements after sales, losses, acquisitions, adjustments and cancellation restocks.	◯	◯	◯	◯	◯
5	Orders keep the variant and price at the time of purchase even when the product is later edited or its price changes.	◯	◯	◯	◯	◯
1.3 Functional appropriateness						
6	Each function achieves its task without unnecessary steps or screens.	◯	◯	◯	◯	◯

Observations and evidence:

2. Performance Efficiency

The degree to which the system performs its functions within specified time and throughput, and uses resources efficiently, under the stated conditions.

No.	Statement	5 SA	4 A	3 N	2 D	1 SD
2.1 Time behaviour						
7	Pages, product searches and order actions respond within acceptable time on typical mobile and Wi-Fi connections.	◯	◯	◯	◯	◯
8	Real-time updates (chat messages, order status, notifications) appear without noticeable delay.	◯	◯	◯	◯	◯
2.2 Resource utilization						
9	The apps use device memory, battery, storage and mobile data reasonably (e.g., compressed images, lazy-loaded pages, paginated lists).	◯	◯	◯	◯	◯
10	Database queries and realtime subscriptions fetch only the data a screen needs.	◯	◯	◯	◯	◯
2.3 Capacity						
11	The system handles the expected number of products, users and concurrent orders (e.g., peak ordering before closing time) without degradation.	◯	◯	◯	◯	◯

Observations and evidence:

3. Compatibility

The degree to which the system can exchange information with other systems and perform its functions while sharing the same environment and resources.

No.	Statement	5 SA	4 A	3 N	2 D	1 SD
3.1 Co-existence						
12	The Android app and the web app run alongside other applications on the same device without conflicts or adverse effects.	◯	◯	◯	◯	◯
3.2 Interoperability						
13	The system exchanges data correctly with its external services: Supabase Auth, Storage and Realtime, and Firebase Cloud Messaging.	◯	◯	◯	◯	◯
14	Reports exported to Excel open correctly and contain the expected data.	◯	◯	◯	◯	◯
15	The web and Android clients show the same data for the same account at the same time.	◯	◯	◯	◯	◯

Observations and evidence:

4. Interaction Capability

The degree to which the intended users can interact with the system to complete their tasks, through its user interface.

No.	Statement	5 SA	4 A	3 N	2 D	1 SD
4.1 Appropriateness recognizability						
16	Users can tell from the interface whether the system fits their needs (e.g., the shop, cart and admin sections are clear from first use).	◯	◯	◯	◯	◯
4.2 Learnability						
17	New customers and administrators can learn the main tasks in a short time without formal training.	◯	◯	◯	◯	◯
4.3 Operability						
18	Controls are easy to operate on both the mobile and desktop layouts.	◯	◯	◯	◯	◯
4.4 User error protection						
19	Input validation, confirmation dialogs, unsaved-changes warnings and insufficient-stock checks protect users from making errors.	◯	◯	◯	◯	◯
4.5 User engagement						
20	The interface is attractive and encourages continued use.	◯	◯	◯	◯	◯
4.6 Inclusivity						
21	The system can be used by people of different ages, backgrounds and levels of digital experience, including farmers and small food businesses.	◯	◯	◯	◯	◯
4.7 User assistance						
22	The system supports users with different abilities (e.g., readable text sizes, colour contrast, light and dark themes, labels for screen readers, adequate touch targets).	◯	◯	◯	◯	◯
4.8 Self-descriptiveness						
23	Status labels, empty states, error messages and notifications make the system's state and next steps clear without a manual.	◯	◯	◯	◯	◯

Observations and evidence:

5. Reliability

The degree to which the system performs its functions without failure under specified conditions for a specified period.

No.	Statement	5 SA	4 A	3 N	2 D	1 SD
5.1 Faultlessness						
24	The system performs its functions without errors or crashes under normal operation.	◯	◯	◯	◯	◯
5.2 Availability						
25	The system is up and reachable whenever customers and administrators need it.	◯	◯	◯	◯	◯
5.3 Fault tolerance						
26	The system keeps working, with clear messages, when the network drops or a request to the backend fails; no partial orders are created.	◯	◯	◯	◯	◯
5.4 Recoverability						
27	After an interruption, the system restores the user's session, cart and data (e.g., the server-side cart, re-login, pull-to-refresh).	◯	◯	◯	◯	◯
28	Data can be restored from backups if the database is lost or corrupted.	◯	◯	◯	◯	◯

Observations and evidence:

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6. Security

The degree to which the system protects information and data so that people and other systems have access appropriate to their authorization, and defends against attack.

No.	Statement	5 SA	4 A	3 N	2 D	1 SD
6.1 Confidentiality						
29	Users can access only data they are authorized to see (e.g., customers see only their own orders, cart and chats; valid-ID images are not publicly accessible).	◯	◯	◯	◯	◯
30	Row-level security policies are enabled on every table that holds user or business data.	◯	◯	◯	◯	◯
6.2 Integrity						
31	Prices, stock, order status and user roles cannot be changed by unauthorized users, including through direct API calls with the public key.	◯	◯	◯	◯	◯
6.3 Non-repudiation						
32	Orders, stock movements and status changes are recorded with timestamps so that they can be proven to have taken place.	◯	◯	◯	◯	◯
6.4 Accountability						
33	Administrative actions (stock adjustments, order status changes, user approvals) can be traced to the account that performed them.	◯	◯	◯	◯	◯
6.5 Authenticity						
34	User identity is verified through authentication, email verification, password reset and administrator approval of new accounts.	◯	◯	◯	◯	◯
6.6 Resistance						
35	The system resists common attacks (e.g., injection, cross-site scripting, brute-force login, exposure of secret keys in the client or repository).	◯	◯	◯	◯	◯

Observations and evidence:

7. Maintainability

The degree to which the system can be modified effectively and efficiently by its intended maintainers.

No.	Statement	5 SA	4 A	3 N	2 D	1 SD
7.1 Modularity						
36	The code is organized into separate parts (pages, components, data hooks, models, route guards) so that a change to one has little effect on others.	◯	◯	◯	◯	◯
7.2 Reusability						
37	Components, hooks and utilities are reused rather than duplicated.	◯	◯	◯	◯	◯
7.3 Analysability						
38	The causes of defects and the effects of a planned change can be found easily (readable code, TypeScript types, linting, README documentation).	◯	◯	◯	◯	◯
7.4 Modifiability						
39	Features can be changed or added without introducing defects or lowering quality.	◯	◯	◯	◯	◯
7.5 Testability						
40	Test criteria can be set and tests run effectively (e.g., automated tests, test data, separate test environment).	◯	◯	◯	◯	◯

Observations and evidence:

8. Flexibility

The degree to which the system can be adapted to changes in its requirements, contexts of use or system environment.

No.	Statement	5 SA	4 A	3 N	2 D	1 SD
8.1 Adaptability						
41	The system works across supported browsers, Android versions and screen sizes.	◯	◯	◯	◯	◯
8.2 Scalability						
42	The system can handle growth in products, users, orders or additional sellers by scaling its resources.	◯	◯	◯	◯	◯
8.3 Installability						
43	The Android APK installs, updates and uninstalls cleanly, and the web app can be built and deployed by following the README.	◯	◯	◯	◯	◯
8.4 Replaceability						
44	The system can replace the current manual, paper or social-media ordering process for the same purpose.	◯	◯	◯	◯	◯

Observations and evidence:

9. Safety

The degree to which the system, under defined conditions, avoids states that put human life, health, property or the environment at risk. For this system, the items address harm to the farmer group's stock, orders, money and records.

No.	Statement	5 SA	4 A	3 N	2 D	1 SD
9.1 Operational constraint						
45	The system keeps operations within safe limits (e.g., no ordering beyond available stock, no negative stock, no ordering outside store hours).	◯	◯	◯	◯	◯
9.2 Risk identification						
46	The system identifies conditions that could cause loss (e.g., low-stock alerts, insufficient-stock detection, orders affected by closing time).	◯	◯	◯	◯	◯
9.3 Fail safe						
47	When an operation fails, the system leaves data in a safe state (e.g., a failed checkout neither creates an order nor deducts stock).	◯	◯	◯	◯	◯
9.4 Hazard warning						
48	The system warns users before actions that are hard to undo (e.g., deleting a product or variant, cancelling an order, adjusting stock, closing with unsaved changes).	◯	◯	◯	◯	◯
9.5 Safe integration						
49	Integrated services (push notifications, edge functions, Excel export) do not expose data or leave the system in an inconsistent state.	◯	◯	◯	◯	◯

Observations and evidence:

Part IV. Overall Assessment

1. What are the strengths of Baybayani as a product?

2. What defects or risks did you find? Give the steps to reproduce, the file or the screen where possible.

3. What do you recommend before wider deployment?

Thank you for your time and expert assessment.